Mazes vs Pathfinding (algorithms)

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***Abstract*—READ CONC AFTER ABSTRACT TO DOUBLE CHECK IF PAPER IS WORTH USING**

This paper demonstrates my experiment to run various paathfinding algorithms against differeing maze generation algorithms.

I. Introduction

This investigation is a 'fun' visual way to compare search algorithms in their speed accuracy when navigating a variety of mazes generated by different algorithms. This should show the strengths and

II. Literature Review

might not need

III. Related work

what was the paper,defined by the author(surname) -¿ summarise the paper, bit you need? The Authors of Analysis of Maze Generation Algorithms[2] a -¿ first to author surname et al

i think mine is a planning around ai as seraching?? ai arent learning anything

IV. Implementation

using (git repos autho) repo (reference) we expanded on this to allow for multiple maze algorithms and pathfinders.

V. The Experiment

can make a perfect maze without any storage annndd if can garantee the choices(e.g with a seed?) don't even need to store whole maze.

VI. The Maze Algorithms

maze algorithms used will be from Jamis Buck's book Mazes for Programmers[1]

section on the maze algorithms/tree/graph gen algorithms I am using will have subsections

A. *binaryTree*

B. *depth first/recursive backtracker*

VII. The Pathfinding Algorithms

section on pathfind/tree/graph search algorithms i am using USE ϵ-GREEDY-¿ with binaryTree due to the bias, compare 'with grain vs against grain' will have subsections

VIII. Results

IX. Conclusion

X. Future Work

References

[1] Jamis Buck. *Mazes for programmers: code your own twisty little passages*. eng. 1st edition. The Pragmatic Programmers. Dallas, Texas ; The Pragmatic Bookshelf, 2015. ISBN: 978-1-68050-131-5.

[2] J. Vijaya et al. "Analysis of Maze Generation Algorithms". In: *2024 5th International Conference on Innovative Trends in Information Technology (ICITIIT)*. Mar. 2024, pp. 1–6. DOI: 10.1109/ICITIIT61487.2024.10580178. URL: https://ieeexplore.ieee.org/document/10580178 (visited on 02/16/2026).